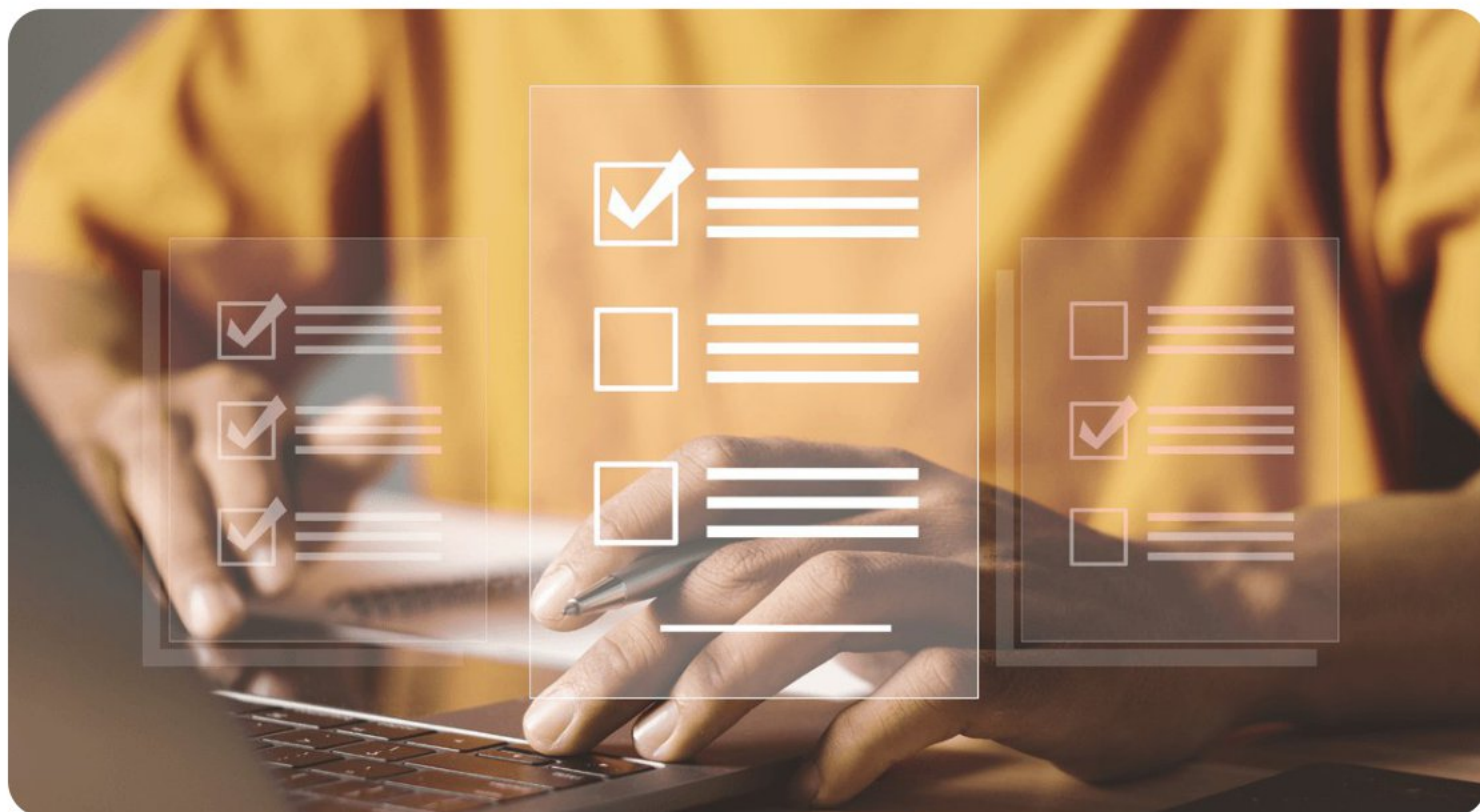




Semana 6

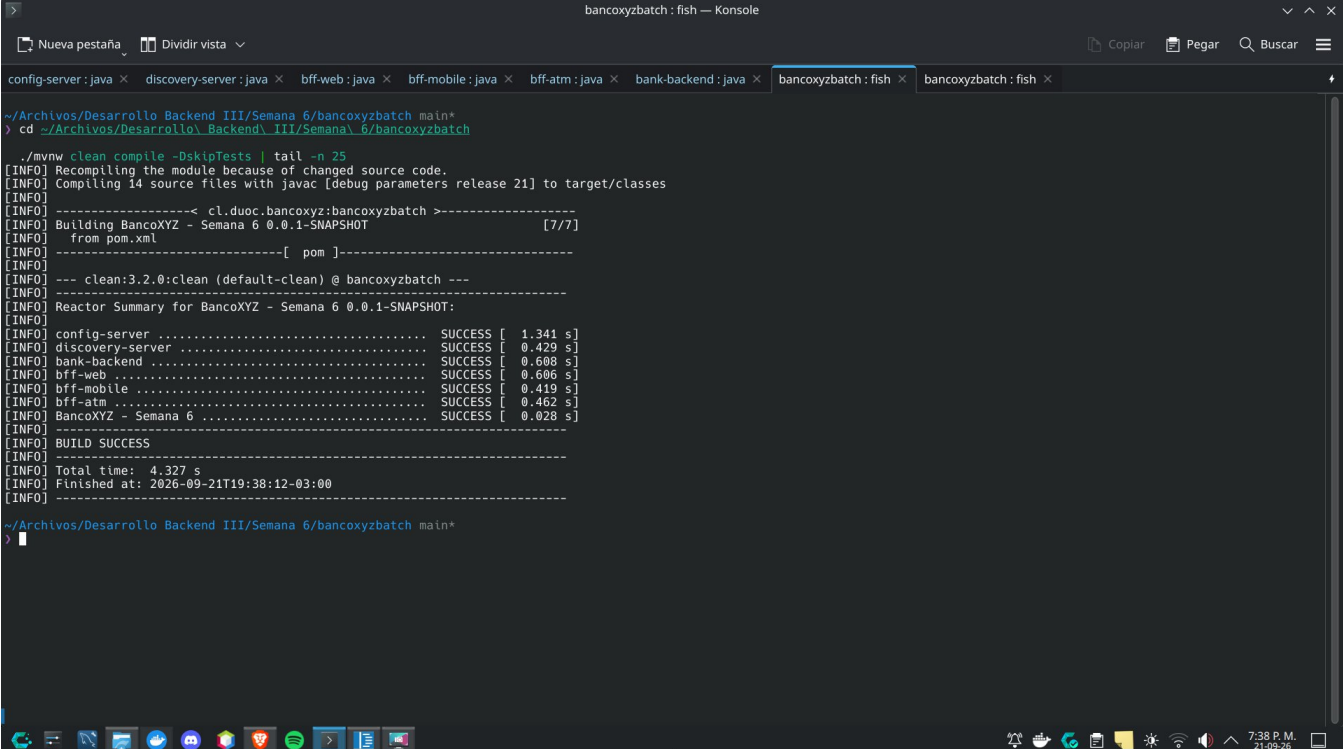
Desarrollo Backend III (PBY2203)

Nombre estudiantes:	Natalia Alvarado
Asignatura: Desarrollo Backend III	Carrera: Analista Programador Computacional
Profesor: Alonso Castillo	Fecha: 21-09-2026

Índice de evidencias

Nº	Evidencia	Pág.
01	Compilación exitosa de la arquitectura completa	3
02	Configuración centralizada mediante Spring Cloud Config Server	4
03	Registro de microservicios en Eureka Service Discovery	5
04	Ejecución de las APIs Web, Mobile y ATM sobre datos migrados	6
05	Autenticación y autorización por canal mediante Bearer Token	7
06	Auditoría integral de Actuator y mecanismos de resiliencia	8
07	Retry de tres intentos ante indisponibilidad del Bank Backend	9
08	Apertura del Circuit Breaker y bloqueo de nuevas solicitudes	10
09	Recuperación automática del Circuit Breaker	11
10	Repositorio GitHub y estructura final	12

01 Compilación exitosa de la arquitectura completa

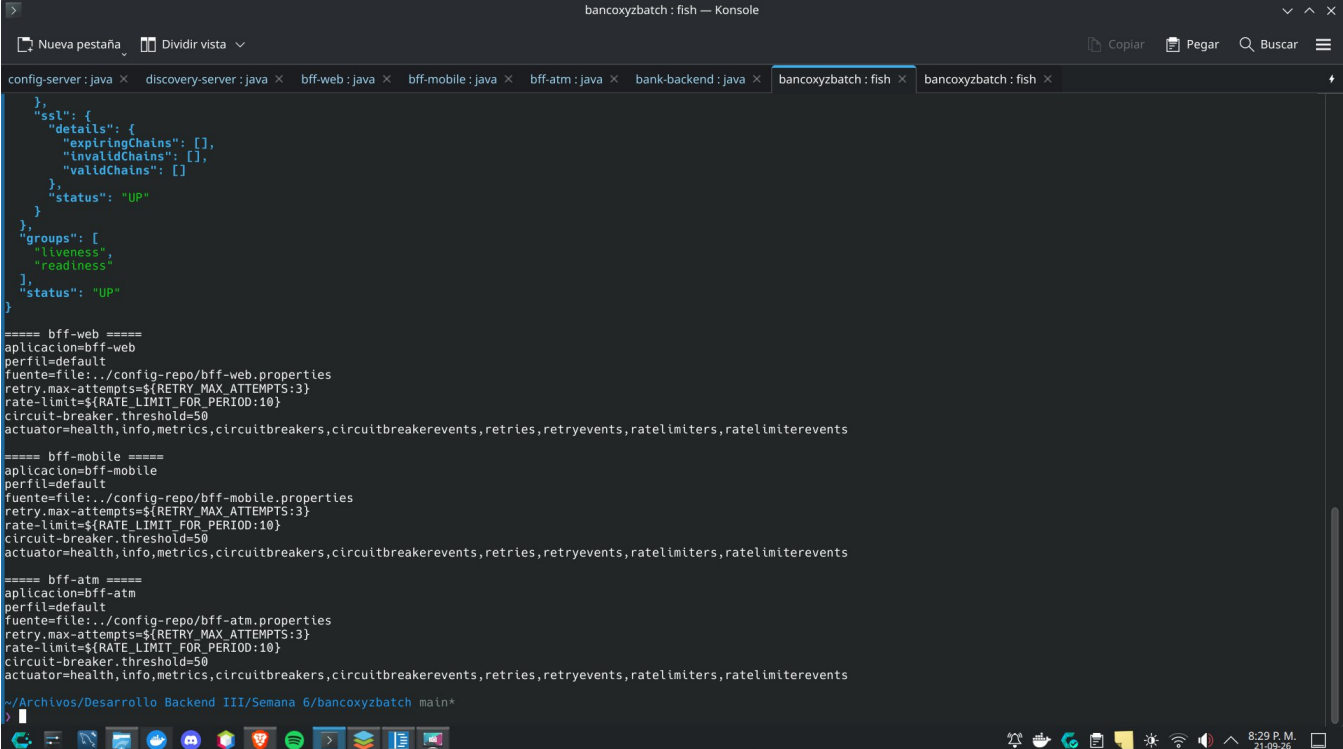


The screenshot shows a terminal window titled "bancoxyzbatch : fish — Konsole". The terminal displays the output of a Maven build command. The build is successful, and the summary shows the following components and their build times:

Component	Status	Time
config-server	SUCCESS	1.341 s
discovery-server	SUCCESS	0.429 s
bank-backend	SUCCESS	0.608 s
bff-web	SUCCESS	0.606 s
bff-mobile	SUCCESS	0.419 s
bff-atm	SUCCESS	0.462 s
BancoXYZ - Semana 6	SUCCESS	0.028 s

The total build time is 4.327 s, and the build finished at 2026-09-21T19:38:12-03:00.

02 Configuración centralizada mediante Spring Cloud Config Server



The screenshot shows a terminal window titled "bancoxyzbatch : fish — Konsole". The terminal displays the configuration for a Spring Cloud Config Server. The configuration is organized into sections for different applications: bff-web, bff-mobile, and bff-atm. Each section includes properties for application name, profile, source, retry attempts, rate limit, circuit breaker threshold, and actuators.

```
config-server: java × discovery-server: java × bff-web: java × bff-mobile: java × bff-atm: java × bank-backend: java × bancoxyzbatch: fish × bancoxyzbatch: fish ×
},
"ssl": {
  "details": {
    "exploringChains": [],
    "invalidChains": [],
    "validChains": []
  },
  "status": "UP"
}
},
"groups": [
  "liveness",
  "readiness"
],
"status": "UP"
}
}

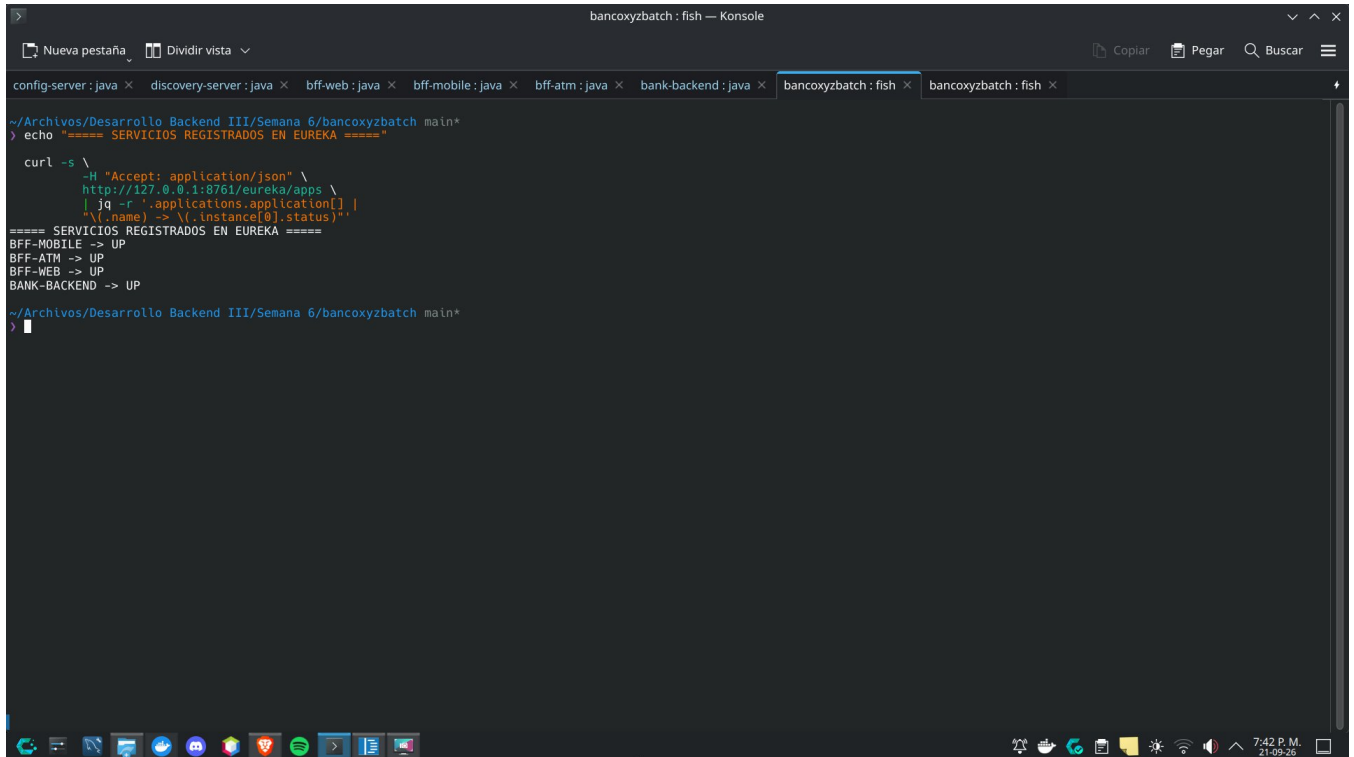
===== bff-web =====
aplicacion=bff-web
perfil=default
fuente=file:../config-repo/bff-web.properties
retry.max-attempts=${RETRY_MAX_ATTEMPTS:3}
rate-limit=${RATE_LIMIT_FOR_PERIOD:10}
circuit-breaker.threshold=50
actuator=health,info,metrics,circuitbreakers,circuitbreakerevents,retries,retryevents,ratelimiters,ratelimiterevents

===== bff-mobile =====
aplicacion=bff-mobile
perfil=default
fuente=file:../config-repo/bff-mobile.properties
retry.max-attempts=${RETRY_MAX_ATTEMPTS:3}
rate-limit=${RATE_LIMIT_FOR_PERIOD:10}
circuit-breaker.threshold=50
actuator=health,info,metrics,circuitbreakers,circuitbreakerevents,retries,retryevents,ratelimiters,ratelimiterevents

===== bff-atm =====
aplicacion=bff-atm
perfil=default
fuente=file:../config-repo/bff-atm.properties
retry.max-attempts=${RETRY_MAX_ATTEMPTS:3}
rate-limit=${RATE_LIMIT_FOR_PERIOD:10}
circuit-breaker.threshold=50
actuator=health,info,metrics,circuitbreakers,circuitbreakerevents,retries,retryevents,ratelimiters,ratelimiterevents

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
}
```

03 Registro de microservicios en Eureka Service Discovery



The screenshot shows a terminal window titled "bancoxyzbatch : fish — Konsole". The terminal displays the following commands and output:

```
~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*  
> echo "==== SERVICIOS REGISTRADOS EN EUREKA ===="
```

The output of the command is:

```
==== SERVICIOS REGISTRADOS EN EUREKA ====  
BFF-MOBILE -> UP  
BFF-ATM -> UP  
BFF-WEB -> UP  
BANK-BACKEND -> UP
```

The terminal window also shows several tabs at the top: "config-server : java", "discovery-server : java", "bff-web : java", "bff-mobile : java", "bff-atm : java", "bank-backend : java", "bancoxyzbatch : fish", and "bancoxyzbatch : fish".

04 Ejecución de las APIs Web, Mobile y ATM sobre datos migrados

```
bancoxyzbatch : fish — Konsole
Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar  ☰
config-server : java ×  discovery-server : java ×  bff-web : java ×  bff-mobile : java ×  bff-atm : java ×  bank-backend : java ×  bancoxyzbatch : fish ×  bancoxyzbatch : fish ×
echo
echo "===== MOBILE ====="
curl -k -s \
  -H "Authorization: Bearer bancoxyz-mobile-demo-token-2026" \
  https://127.0.0.1:8082/api/mobile/cuentas \
  | jq '.[0]'

echo
echo "===== ATM ====="
curl -k -s \
  -H "Authorization: Bearer bancoxyz-atm-demo-token-2026" \
  https://127.0.0.1:8083/api/atm/cuentas/101/saldo \
  | jq

===== WEB =====
{
  "activo": true,
  "cuentaId": 101,
  "estado": "PROCESADO",
  "interesCalculado": 50.0,
  "nombre": "John Doe",
  "observacion": "Interés mensual calculado con tasa de ahorro del 1%",
  "saldoFinal": 5050.0,
  "saldoInicial": 5000.0,
  "tasaInteres": 0.01,
  "tipo": "ahorro",
  "ultimaInstanciaId": 19
}

===== MOBILE =====
{
  "cuentaId": 101,
  "nombre": "John Doe",
  "tipo": "ahorro",
  "estado": "PROCESADO",
  "saldoFinal": 5050.0
}

===== ATM =====
{
  "cuentaId": 101,
  "nombre": "John Doe",
  "saldoDisponible": 5050.0
}

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
```

05 Autenticación y autorización por canal mediante Bearer Token

```
bancoxyzbatch : fish — Konsole

Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar  ☰

config-server:java ×  discovery-server:java ×  bff-web:java ×  bff-mobile:java ×  bff-atm:java ×  bank-backend:java ×  bancoxyzbatch:fish ×  bancoxyzbatch:fish ×

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
> echo "==== AUTENTICACION ====="

curl -k -s -o /dev/null \
  -w "WEB sin token -> HTTP %{http_code}\n" \
  https://127.0.0.1:8081/api/web/cuentas

echo
echo "==== AUTORIZACION CRUZADA ====="

curl -k -s -o /dev/null \
  -w "WEB token -> MOBILE -> HTTP %{http_code}\n" \
  -H "Authorization: Bearer bancoxyz-web-demo-token-2026" \
  https://127.0.0.1:8082/api/mobile/cuentas

curl -k -s -o /dev/null \
  -w "MOBILE token -> ATM -> HTTP %{http_code}\n" \
  -H "Authorization: Bearer bancoxyz-mobile-demo-token-2026" \
  https://127.0.0.1:8083/api/atm/cuentas/101/saldo

curl -k -s -o /dev/null \
  -w "ATM token -> WEB -> HTTP %{http_code}\n" \
  -H "Authorization: Bearer bancoxyz-atm-demo-token-2026" \
  https://127.0.0.1:8081/api/web/cuentas

===== AUTENTICACION =====
WEB sin token -> HTTP 401

===== AUTORIZACION CRUZADA =====
WEB token -> MOBILE -> HTTP 403
MOBILE token -> ATM -> HTTP 403
ATM token -> WEB -> HTTP 403

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
> |
```

06 Auditoría integral de Actuator y mecanismos de resiliencia

```
bancoxyzbatch : fish — Konsole
Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar

config-server:java x discovery-server:java x bff-web:java x bff-mobile:java x bff-atm:java x bank-backend:java x bancoxyzbatch:fish x bancoxyzbatch:fish x

BANCOXYZ - AUDITORIA ACTUATOR + RESILIENCE4J - SEMANA 6
Modo: no destructivo
Instancia Resilience4j esperada: bankbackend

===== 1. ACTUATOR HEALTH =====
✓ WEB health -> HTTP 200 / UP
✓ MOBILE health -> HTTP 200 / UP
✓ ATM health -> HTTP 200 / UP

===== 2. CIRCUIT BREAKER REGISTRADO =====
✓ WEB -> bankbackend registrado / estado CLOSED
✓ MOBILE -> bankbackend registrado / estado CLOSED
✓ ATM -> bankbackend registrado / estado CLOSED

===== 3. RETRY REGISTRADO =====
✓ WEB Retry -> bankbackend registrado
✓ MOBILE Retry -> bankbackend registrado
✓ ATM Retry -> bankbackend registrado

===== 4. RATE LIMITER REGISTRADO =====
✓ WEB Rate Limiter -> bankbackend registrado
✓ MOBILE Rate Limiter -> bankbackend registrado
✓ ATM Rate Limiter -> bankbackend registrado

===== 5. APIS PRINCIPALES =====
✓ WEB /api/web/cuentas -> HTTP 200
✓ MOBILE /api/mobile/cuentas -> HTTP 200
✓ ATM /api/atm/cuentas/101/saldo -> HTTP 200

===== 6. RATE LIMITER - RAFAGA CONTROLADA =====
WEB -> 200: 10 | 429: 20 | otros: 0
✓ WEB Rate Limiter funcional
MOBILE -> 200: 10 | 429: 20 | otros: 0
✓ MOBILE Rate Limiter funcional
ATM -> 200: 10 | 429: 20 | otros: 0
✓ ATM Rate Limiter funcional

===== 7. AISLAMIENTO RATE LIMITER / CIRCUIT BREAKER =====
WEB Circuit Breaker después del Rate Limiter -> state=CLOSED | failedCalls=0
✓ Los HTTP 429 no contaminan el Circuit Breaker

=====
RESULTADO: TODAS LAS PRUEBAS NO DESTRUCTIVAS PASARON
Actuator + Circuit Breaker + Retry + Rate Limiter: OK
=====
```

07 Retry de tres intentos ante indisponibilidad del Bank Backend

```
bancoxyzbatch : fish — Konsole

Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar

config-server:java x discovery-server:java x bff-web:java x bff-mobile:java x bff-atm:java x bank-backend:fish x bancoxyzbatch:fish x bancoxyzbatch:fish x

/api/atm/cuentas/101/saldo

echo "===== RETRY -- BANK BACKEND DOWN ====="

for i in (seq 1 3)
  set NAME $NAMES[$i]
  set PORT $PORTS[$i]
  set TOKEN $TOKENS[$i]
  set PATH_API $PATHS[$i]

  echo
  echo "===== $NAME ====="

  curl -k -s -o /dev/null \
    -w "%{http_code}\n" \
    -H "Authorization: Bearer $TOKEN" \
    "https://127.0.0.1:$PORT$PATH_API"

  curl -k -s \
    -H "Authorization: Bearer $TOKEN" \
    "https://127.0.0.1:$PORT/actuator/retryevents" \
    | jq -r '.retryEvents[-3:][0] |
    "intento=\(.numberOfAttempts) tipo=\(.type)'"
end
===== RETRY -- BANK BACKEND DOWN =====

===== WEB =====
WEB -> HTTP 503
intento=1 tipo=RETRY
intento=2 tipo=RETRY
intento=3 tipo=ERROR

===== MOBILE =====
MOBILE -> HTTP 503
intento=1 tipo=RETRY
intento=2 tipo=RETRY
intento=3 tipo=ERROR

===== ATM =====
ATM -> HTTP 503
intento=1 tipo=RETRY
intento=2 tipo=RETRY
intento=3 tipo=ERROR

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
>
```

```
bancoxyzbatch : fish — Konsole

Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar

config-server:java x discovery-server:java x bff-web:java x bff-mobile:java x bff-atm:java x bank-backend:fish x bancoxyzbatch:fish x bancoxyzbatch:fish x

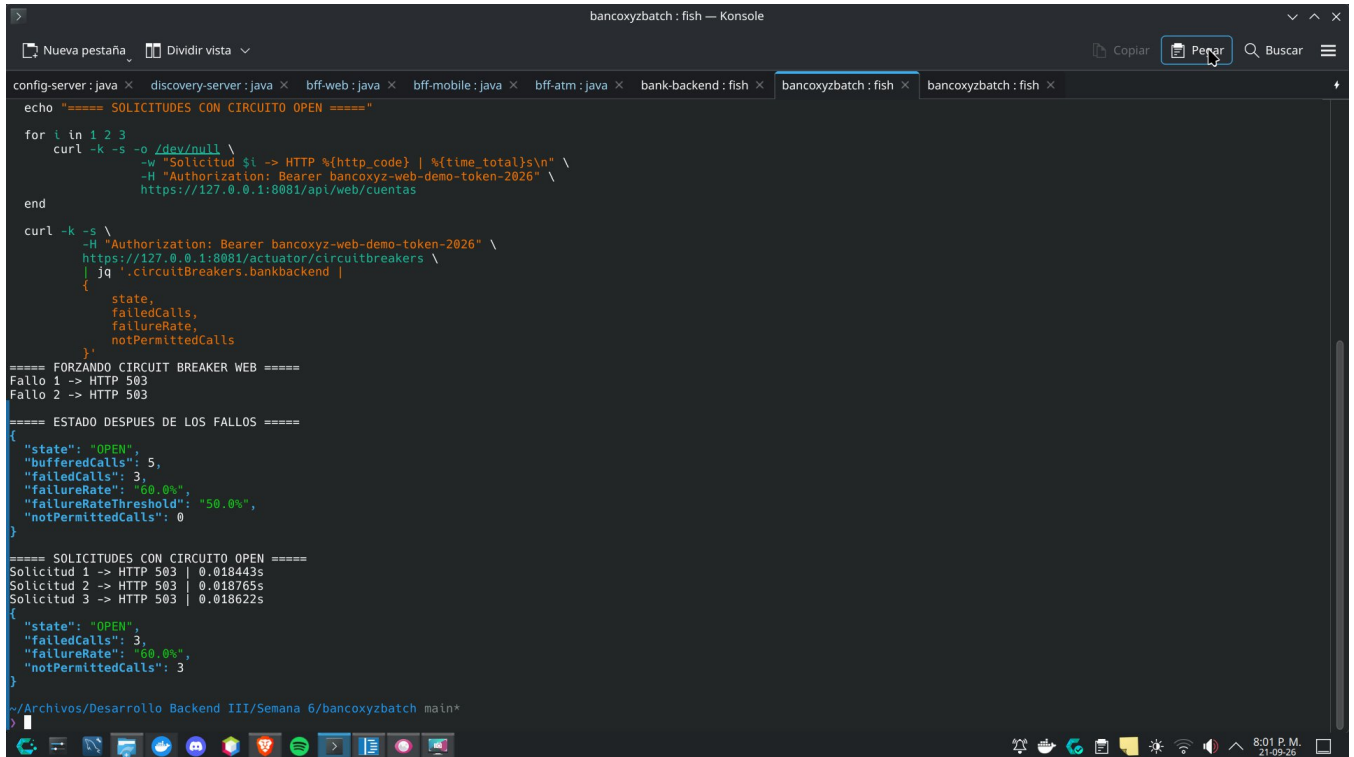
~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
> ss -ltnp | grep -E '(:8080|:8761|:8080|:8081|:8082|:8083)'
LISTEN 0      100          *:8088      *:*        users:((("java",pid=267668,fd=75))
LISTEN 0      100          *:8761      *:*        users:((("java",pid=268158,fd=129))
LISTEN 0      100          *:8083      *:*        users:((("java",pid=295837,fd=161))
LISTEN 0      100          *:8082      *:*        users:((("java",pid=294747,fd=164))
LISTEN 0      100          *:8081      *:*        users:((("java",pid=294459,fd=164))
LISTEN 0      100          *:8080      *:*        users:((("java",pid=302277,fd=145))

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
> fuser -k 8080/tcp
sleep 2
8080/tcp:      302277

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
> if ss -ltnp | grep -q ':8080'
  echo "X BANK-BACKEND SIGUE VIVO"
else
  echo "X BANK-BACKEND DOWN"
end
X BANK-BACKEND DOWN

~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
>
```

08 Apertura del Circuit Breaker y bloqueo de nuevas solicitudes



```
config-server: java × discovery-server: java × bff-web: java × bff-mobile: java × bff-atm: java × bank-backend: fish × bancoxyzbatch: fish × bancoxyzbatch: fish ×
echo "===== SOLICITUDES CON CIRCUITO OPEN ====="

for i in 1 2 3
  curl -k -s -o /dev/null \
    -w "Solicitud $i -> HTTP %{http_code} | %{time_total}s\n" \
    -H "Authorization: Bearer bancoxyz-web-demo-token-2026" \
    https://127.0.0.1:8081/api/web/cuentas
end

curl -k -s \
  -H "Authorization: Bearer bancoxyz-web-demo-token-2026" \
  https://127.0.0.1:8081/actuator/circuitbreakers \
  | jq '.circuitBreakers.bankbackend |
  {
    state,
    failedCalls,
    failureRate,
    notPermittedCalls
  }'

===== FORZANDO CIRCUITO BREAKER WEB =====
Fallo 1 -> HTTP 503
Fallo 2 -> HTTP 503

===== ESTADO DESPUES DE LOS FALLOS =====
{
  "state": "OPEN",
  "bufferedCalls": 5,
  "failedCalls": 3,
  "failureRate": "60.0%",
  "failureRateThreshold": "50.0%",
  "notPermittedCalls": 0
}

===== SOLICITUDES CON CIRCUITO OPEN =====
Solicitud 1 -> HTTP 503 | 0.018443s
Solicitud 2 -> HTTP 503 | 0.018765s
Solicitud 3 -> HTTP 503 | 0.018622s
{
  "state": "OPEN",
  "failedCalls": 3,
  "failureRate": "60.0%",
  "notPermittedCalls": 3
}

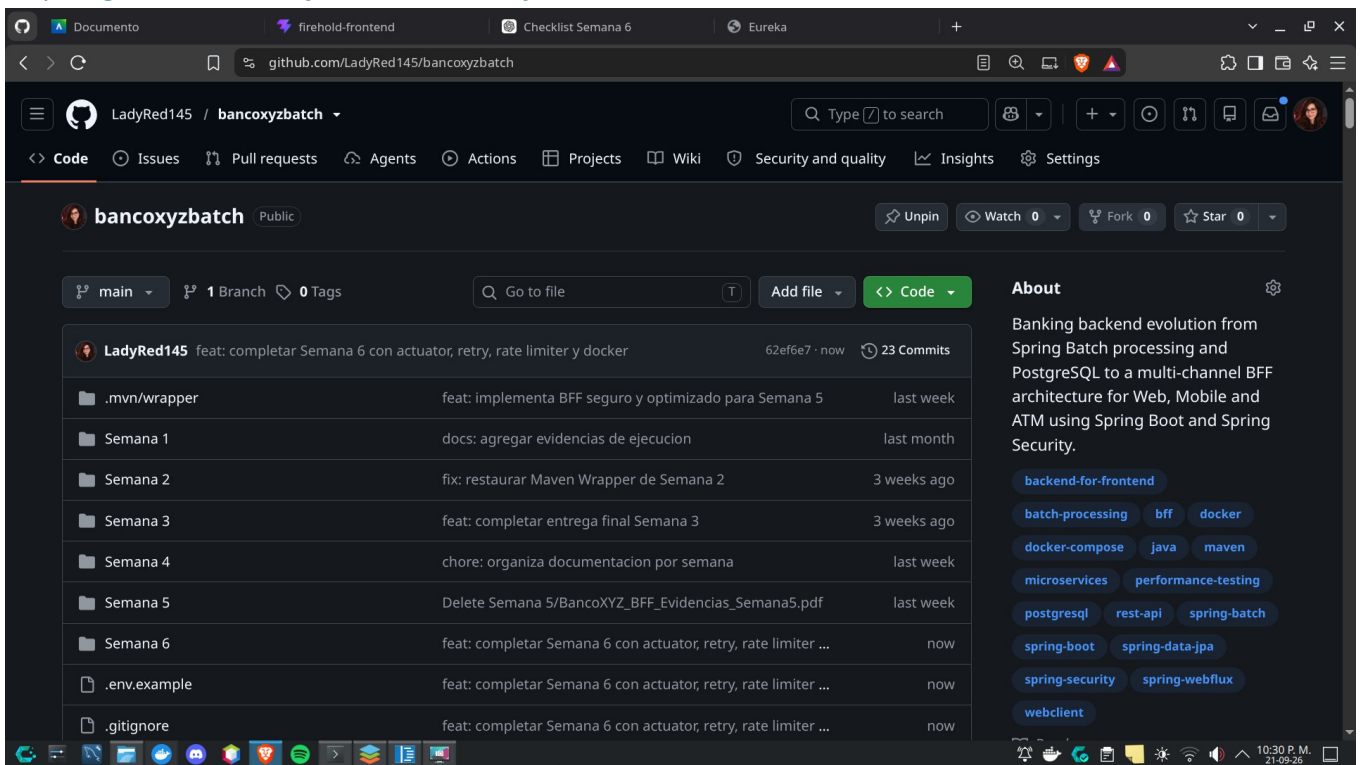
~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main*
```

09 Recuperación automática del Circuit Breaker

```
bancoxyzbatch : fish — Konsole
Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar  ☰
config-server:java × discovery-server:java × bff-web:java × bff-mobile:java × bff-atm:java × bank-backend:java × bancoxyzbatch:fish × bancoxyzbatch:fish ×
for i in 1 2
  curl -k -s -o /dev/null \
    -w "Probe $i -> HTTP %{http_code}\n" \
    -H "Authorization: Bearer bancoxyz-web-demo-token-2026" \
    https://127.0.0.1:8081/api/web/cuentas
end
echo
echo "===== ESTADO FINAL ====="
curl -k -s \
  -H "Authorization: Bearer bancoxyz-web-demo-token-2026" \
  https://127.0.0.1:8081/actuator/circuitbreakers \
  | jq '.circuitBreakers.bankbackend |
  {
    state,
    failedCalls,
    failureRate,
    notPermittedCalls
  }'
===== BANK BACKEND RECUPERADO =====
LISTEN 0      100          *:8080       *: *    users:((("java",pid=337762,fd=145))
Esperando transición OPEN -> HALF_OPEN...
===== ESTADO HALF_OPEN =====
{
  "state": "HALF_OPEN",
  "failedCalls": 0,
  "notPermittedCalls": 0
}
===== PROBES DE RECUPERACION =====
Probe 1 -> HTTP 200
Probe 2 -> HTTP 200
===== ESTADO FINAL =====
{
  "state": "CLOSED",
  "failedCalls": 0,
  "failureRate": "-1.0%",
  "notPermittedCalls": 0
}
~/Archivos/Desarrollo Backend III/Semana 6/bancoxyzbatch main* 11s
```

10 Repositorio GitHub y estructura final

<https://github.com/LadyRed145/bancoxyzbatch/tree/main/Semana%206>



The screenshot displays the GitHub repository page for **LadyRed145 / bancoxyzbatch**. The repository is public and has 0 forks, 0 stars, and 0 watchers. The main branch is **main** with 1 branch and 0 tags. The repository contains 23 commits.

The repository structure is as follows:

File/Folder	Description	Last Commit
LadyRed145	feat: completar Semana 6 con actuador, retry, rate limiter y docker	62ef6e7 · now
.mvn/wrapper	feat: implementa BFF seguro y optimizado para Semana 5	last week
Semana 1	docs: agregar evidencias de ejecucion	last month
Semana 2	fix: restaurar Maven Wrapper de Semana 2	3 weeks ago
Semana 3	feat: completar entrega final Semana 3	3 weeks ago
Semana 4	chore: organiza documentacion por semana	last week
Semana 5	Delete Semana 5/BancoXYZ_BFF_Evidencias_Semana5.pdf	last week
Semana 6	feat: completar Semana 6 con actuador, retry, rate limiter ...	now
.env.example	feat: completar Semana 6 con actuador, retry, rate limiter ...	now
.gitignore	feat: completar Semana 6 con actuador, retry, rate limiter ...	now

The **About** section on the right describes the project as a banking backend evolution from Spring Batch processing and PostgreSQL to a multi-channel BFF architecture for Web, Mobile and ATM using Spring Boot and Spring Security. The tags listed are: **backend-for-frontend**, **batch-processing**, **bff**, **docker**, **docker-compose**, **java**, **maven**, **microservices**, **performance-testing**, **postgresql**, **rest-api**, **spring-batch**, **spring-boot**, **spring-data-jpa**, **spring-security**, **spring-webflux**, and **webclient**.



Reservados todos los derechos Fundación Instituto Profesional Duoc UC. No se permite copiar, reproducir, reeditar, descargar, publicar, emitir, difundir, de forma total o parcial la presente obra, ni su incorporación a un sistema informático, ni su transmisión en cualquier forma o por cualquier medio (electrónico, mecánico, fotocopia, grabación u otros) sin autorización previa y por escrito de Fundación Instituto Profesional Duoc UC. La infracción de dichos derechos puede constituir un delito contra la propiedad intelectual.